Supply Chain Performance Analysis — Insights

1. Key Performance Metrics

KPI	Value
Total Production Volume	57K
Total Products Sold	46K
Total Revenue	577.60K
Total Cost	52.92K

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2. Overall View

Overall Insights:

- The analysis covers a total production volume of approximately 57K and 46K products sold, generating 577.60K total revenue.
- Supplier 1 is the highest revenue-generating supplier, contributing approximately 157.53K.
- Skincare generates the highest revenue at approximately 241.63K, but it also has the highest manufacturing cost and second highest defect rate value.
- Road transportation has the highest transportation cost at approximately 16.0K, while Sea has the lowest at approximately 7.1K.
- The Inspection Result Distribution shows that Pending inspections account for the largest share at 41%, followed by Fail at 36% and Pass at 23%.

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3. Chart-wise Insights + Suggestions

3.1 Revenue by Supplier

Insights:

- Supplier 1 generates the highest revenue at approximately 157.53K.
- The revenue contribution varies across suppliers, indicating differences in supplier performance.

Suggestions:

- Analyse the factors contributing to Supplier 1's higher revenue and identify practices that can be applied to other suppliers.
- Review the performance of lower-revenue suppliers and evaluate opportunities to improve their contribution.
- Monitor supplier performance regularly to support better supplier-management decisions.

3.2 Revenue by Product Type

Insights:

- Skincare has a noticeably higher revenue contribution compared with Haircare and Cosmetics.

Suggestions:

- Maintain sufficient inventory availability for Skincare due to its higher revenue contribution.
- Analyse the factors behind Skincare's stronger performance, such as demand and product availability.
- Identify opportunities to increase the revenue contribution of Haircare and Cosmetics.

3.3 Manufacturing Cost by Product Type

Insights:

- Skincare has both the highest revenue and the highest manufacturing cost among the three product types.

Suggestions:

- Analyse whether the higher manufacturing cost of Skincare is aligned with its higher revenue contribution.
- Review manufacturing processes for high-cost product types to identify possible efficiency improvements.

3.4 Defect Rate by Product Type

Insights:

- Haircare has the highest defect-rate metric at approximately 2.5, followed by Skincare (2.3), while Cosmetics has the lowest value at approximately 1.9.
- The difference between product types indicates that quality-related performance varies considerably across products.

Suggestions:

- Investigate the reasons behind the higher defect-rate values for Skincare and Haircare.
- Review manufacturing and quality-control processes for higher-defect product types.

3.5 Inspection Result Distribution

Insights:

- Pending inspections form the largest share at 41%, followed by Fail results at 36%, while only 23% of inspections are marked as Pass.
- The relatively high proportion of Pending and Fail results indicates that inspection outcomes require attention.

Suggestions:

- Investigate why a large proportion of inspections remain Pending.
- Analyse the causes of failed inspections and identify recurring quality issues.
- Improve inspection follow-up and quality-control processes to increase successful inspection outcomes.

3.6 Total Cost by Transportation Mode

Insights:

- Road, Rail and Air have relatively similar costs, while Sea has a considerably lower transportation cost.

Suggestions:

- Evaluate the suitability of Sea transportation for shipments where lower transportation cost is more important than speed.

4. Final Business Suggestions

1. Focus on Skincare Manufacturing and Quality

Skincare generates the highest revenue, but it also has the highest manufacturing cost and second highest defect rate value. Since Skincare is the strongest revenue-generating product type, focus on improving its manufacturing process and identify ways to reduce its defect rate while maintaining its revenue performance.

2. Evaluate Lower-Cost Transportation Modes

Road transportation has the highest transportation cost, while Sea transportation has the lowest. Evaluate whether suitable shipments can be shifted from Road to Sea transportation to reduce overall transportation costs, wherever shipping requirements allow.

3. Learn from the Highest-Performing Supplier

Supplier 1 generates the highest revenue among all suppliers. Investigate the practices, processes, and performance factors contributing to Supplier 1's stronger performance and consider applying relevant practices to other suppliers to improve their revenue contribution.

4. Improve Inspection Performance

Pending inspections account for the largest share of inspection results, followed by failed inspections. Investigate the reasons for pending and failed inspections and improve the inspection and quality-control process to increase successful inspection outcomes.